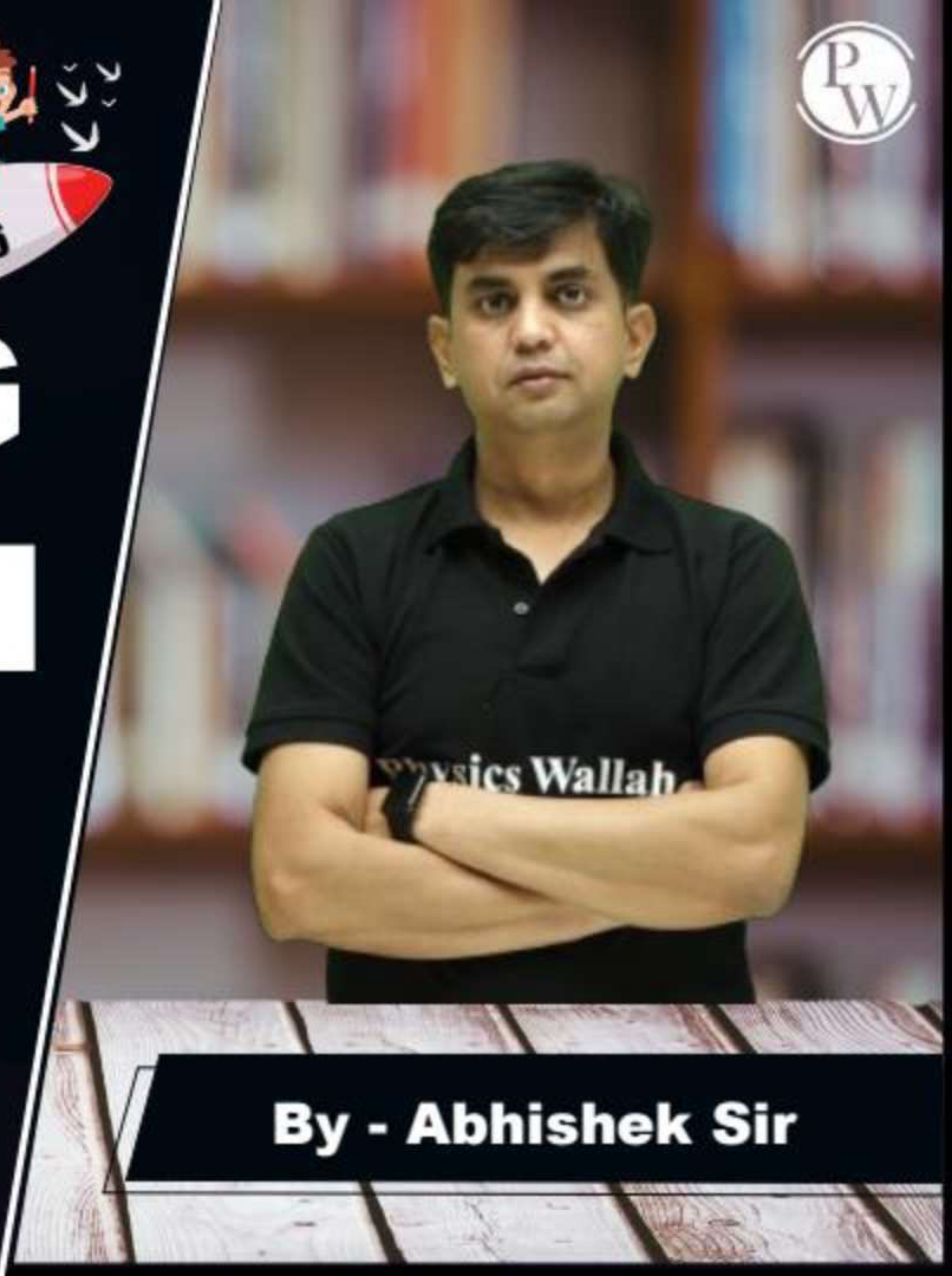
CS & IT ENGINEERING

Computer Network

IPv4 Address

Lecture No. - 09



By - Abhishek Sir

Recap of Previous Lecture

Topic Routing Table

Topics to be Covered

Topic Routing Table

Topic CIDR

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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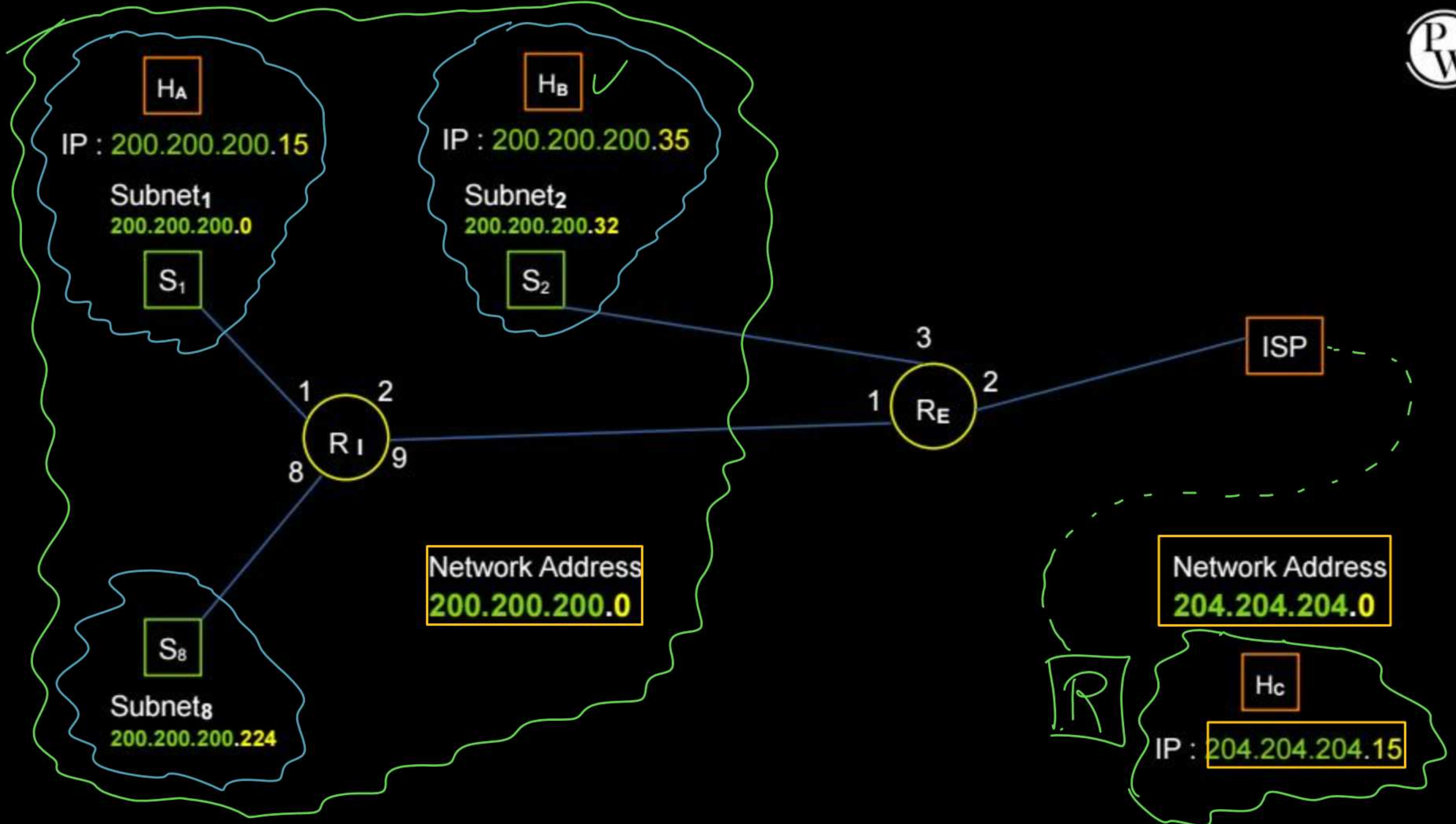
Class C Network :-

Network Address	:	200 . 200 . 200 . 0
Default Netmask	:	255 . 255 . 255 . 0

After 3-bit subnetting :-

Subnet Mask	:	255 . 255 . 255 . 224

1st Subnet Address	:	200 . 200 . 200 . 0
2nd Subnet Address	:	200 . 200 . 200 . 32
3rd Subnet Address	:	200 . 200 . 200 . 64
4th Subnet Address	:	200 . 200 . 200 . 96
5th Subnet Address	:	200 . 200 . 200 . 128
6th Subnet Address	:	200 . 200 . 200 . 160
7th Subnet Address	:	200 . 200 . 200 . 192
8th Subnet Address	:	200 . 200 . 200 . 224



H_A
IP : 200.200.200.15
Subnet1
200.200.200.0
S1
H_B
IP : 200.200.200.35
Subnet2
200.200.200.32
S2
R I
1
2
8
9
S8
Subnet8
200.200.200.224
Network Address
200.200.200.0
R_E
1
2
3
ISP
Network Address
204.204.204.0
R
H_C
IP : 204.204.204.15

Topic : Forwarding Table

Router (R_E) Updated forwarding table

Network Address	Network Mask	Interface ID	Next Hop
200 . 200 . 200 . 0	255 . 255 . 255 . 0	1	R_I
255 . 200 . 200 . 32	255 . 255 . 255 . 224	3	On Link
0 . 0 . 0 . 0 [Default]	0 . 0 . 0 . 0	2	ISP

Router (R_1) Updated forwarding table

Subnet Address	Subnet Mask	Interface ID	Next Hop
200 . 200 . 200 . 0	255 . 255 . 255 . 224	1	On Link
200 . 200 . 200 . 64	255 . 255 . 255 . 224	3	On Link
200 . 200 . 200 . 224	255 . 255 . 255 . 224	8	On Link
Default 0.0.0.0	0.0.0.0	9	R_E

Topic : Forwarding Table

RE

CASE IV :

Source IP Address : 204 . 204 . 204 . 15	HC
Destination IP Address : 200 . 200 . 200 . 35	HB

Router	Interface ID	Next Hop
R_E	3	ON LINK

IP Address : 200 . 200 . 200 . 35
Net Mask : 255 . 255 . 255 . 0

Result ← 200 . 200 . 200 . 0

IP Address : 200 . 200 . 200 . 35
Subnet Mask : 255 . 255 . 255 . 224

Result ← 200 . 200 . 200 . 32

IP Address : 200 . 200 . 200 . 35
Mask : 0 . 0 . 0 . 0

Result ← 0 . 0 . 0 . 0

Topic : Longest Prefix Match

→ Router chooses more specific option over generic one

→ Router chooses the matched entry in which more number of ones in netmask / sub-netmask

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* The default entry in a router's routing table is optional.

* Whenever no any entry matched and no any default entry then router discard the IP datagram and generate ICMP Error.

[MCQ] [GATE-IT-2006] [2 Mark]

IIT-KGP

#Q. A router uses following routing table :

Destination	Mask	Interface	
144.16.0.0	255.255.0.0	Eth0	→ Matched
144.16.64.0	255.255.224.0	Eth1	→ Matched
144.16.68.0	255.255.255.0	Eth2	→ Matched
144.16.68.64	255.255.255.224	Eth3	→ Not matched

A packet bearing a destination address $\boxed{144.16.68.117}$ arrive at the router. On which interface will it be forwarded ?

(A) Eth0

(B) Eth1

(C) Eth2

(D) Eth3

$\boxed{\text{Ans: C}}$

144. 16 . 68.117

255.255.255.224

$\boxed{144.\ 16\ .\ 68.\ 96}$ = Result

117 → 01110101

224 → 11100000

96 ← 01100000

Not matched

144. 16 . 68.117

255.255.255. 0

$\boxed{144.\ 16\ .\ 68.\ \ 0}$ = Result

Matched

[MCQ] H.W. [GATE-CS-2004] [2 Mark]

IIT-D

#Q. The routing table of a router shown below :

Destination	Subnet Mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.255	Eth3
Default		Eth2

On which interfaces will the router forward packets addressed to destinations 128.75.43.16 and 192.12.17.10 respectively ?

(A) Eth1 and Eth2

(B) Eth0 and Eth2

(C) Eth0 and Eth3

(D) Eth1 and Eth3

[NAT] H.W. [GATE-2023][2 Mark] IIT-K

#Q. The forwarding table of a router is shown below.

Subnet Number	Subnet Mask	Interface ID
200.150.0.0	255.255.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3
200.150.68.64	255.255.255.240	4
Default		0

A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID ________.

[NAT] IITKGP [GATE-2014][2 Mark]

#Q. An IP router implementing Classless Inter-domain Routing (CIDR) receives a packet with address 131 . 23 . 151 . 76 . The router's routing table has the following entries :

Prefix	Output Interface
131 . 16 . 0 . 0 / 12	3
131 . 28 . 0 . 0 / 14	5
131 . 19 . 0 . 0 / 16	2
131 . 22 . 0 . 0 / 15	1

The identifier of the output interface on which this packet will be forwarded is ____.

$\boxed{Ans = 1}$

131 . 23 . 151 . 76
255 . 255 . 0 . 0
---
131 . $\boxed{23}$. 0 . 0

Not matched

131 . 23 . 151 . 76
255 . 254 . 0 . 0
---
131 . 22 . 0 . 0

Matched

$23 \rightarrow 00010111$
$254 \rightarrow 11111110$
$22 \leftarrow 00010110$

[MCQ] H.W. [IT-R] [GATE-2025][2 Mark]

#Q. A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

Subnet	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	E1
145.36.128.0	/17	E2
145.36.64.0	/18	E3
145.36.225.0	/24	E4
Default	--	E5

A. E3

B. E1

C. E2

D. E5

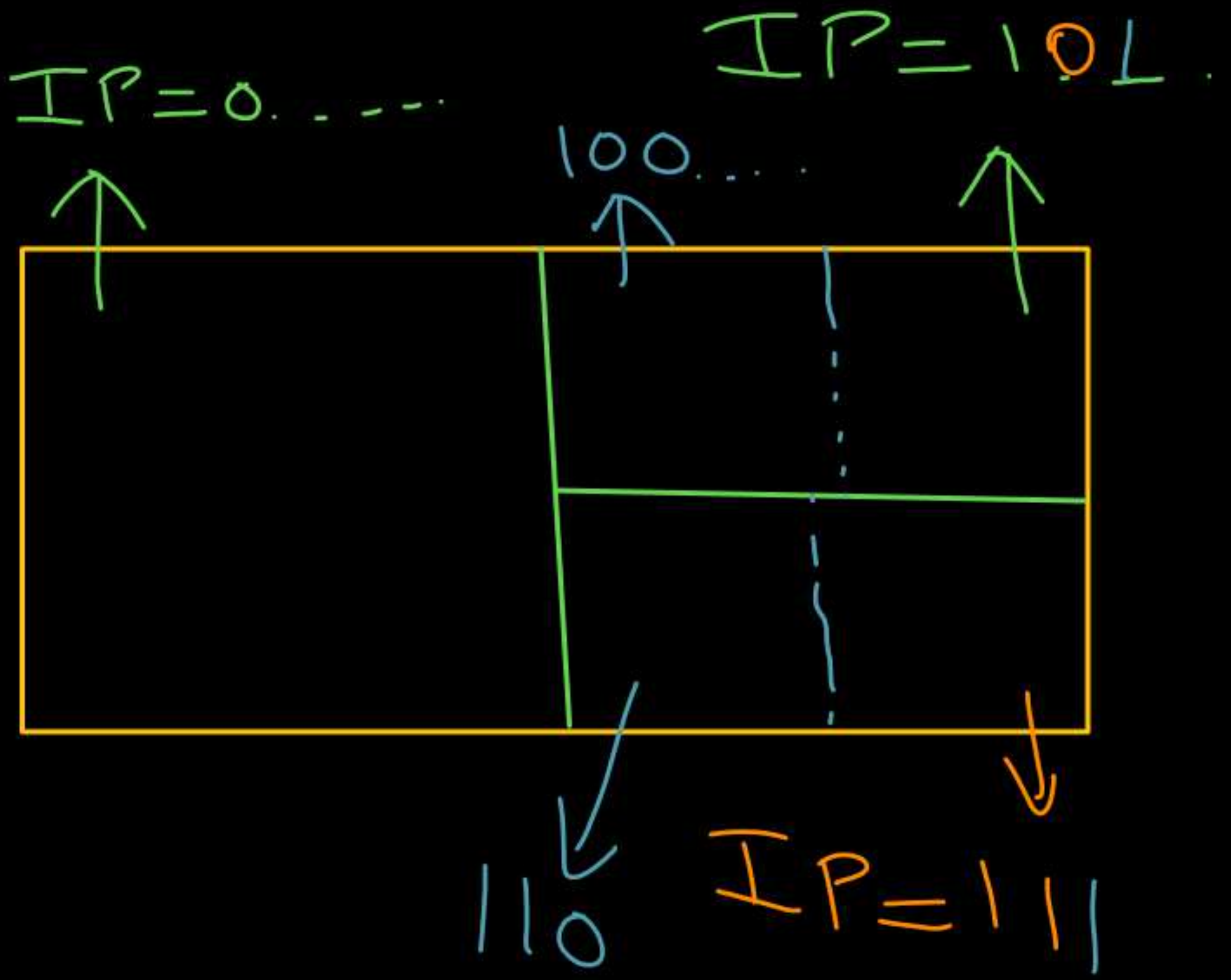
Topic : CIDR

→ CIDR : Classless Inter-Domain Routing

→ IP Address allocation method for IP routing

→ Based on Variable-length subnet masking (VLSM)

→ Allows flexibility in creating 'supernets'



Example :-

[MSQ]

#Q. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it : "155 . 220 . 195 . 0 / 24". An organization request to ISP for range of IP address for its 30 hosts. Which of the following is/are can be a valid (network address) allocation?

(A) ~~155 . 220 . 195 . 144 / 27~~

(B) 155 . 220 . 195 . 160 / 27

(C) 155 . 220 . 195 . 192 / 27

(D) ~~155 . 220 . 195 . 200 / 27~~

ISP having $\boxed{2^8}$ IP Add. Available.

155.220.195.0 to 155.220.195.255

144 → 100 10000

192 → 110 00000

160 → 101 00000

200 → 110 01000

$\boxed{\text{Ans: B\&C}}$

Available IP Address space = $\boxed{155.220.195.0 / 24}$

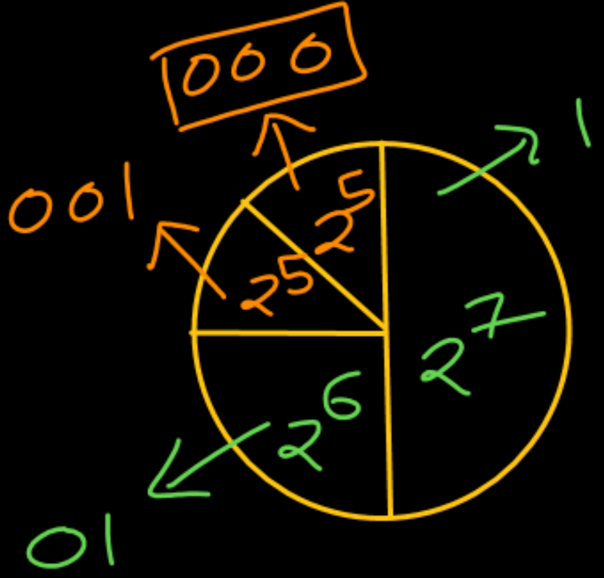
$\underbrace{155.220.195}_{\text{Prefix (24 bit)}}.\underbrace{\_\ \_\ \_\ \_\ \_\ \_\ \_\ \_}_{\text{8 bit Host ID}} / 24$

Number of hosts = 30
[for IP address requested]

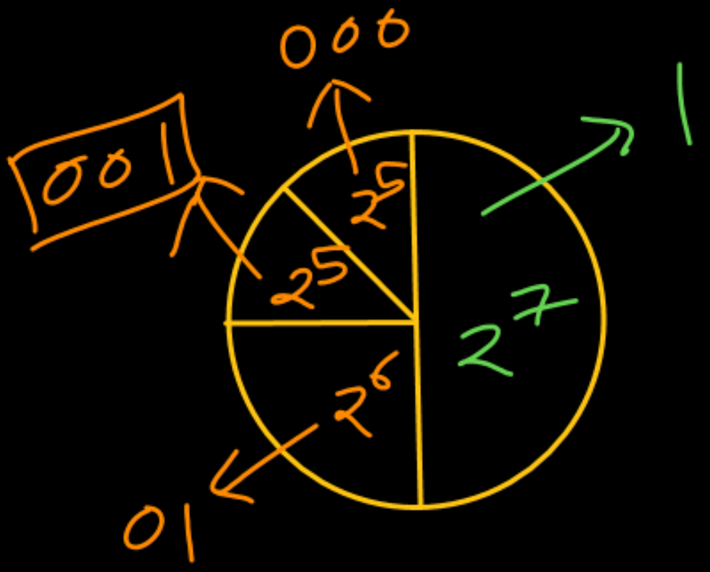
Number of bits for host id in allocated chunk = $\lceil \log_2(\text{Number of hosts}) \rceil$ bits

$= \lceil \log_2(30) \rceil = 5$ bits

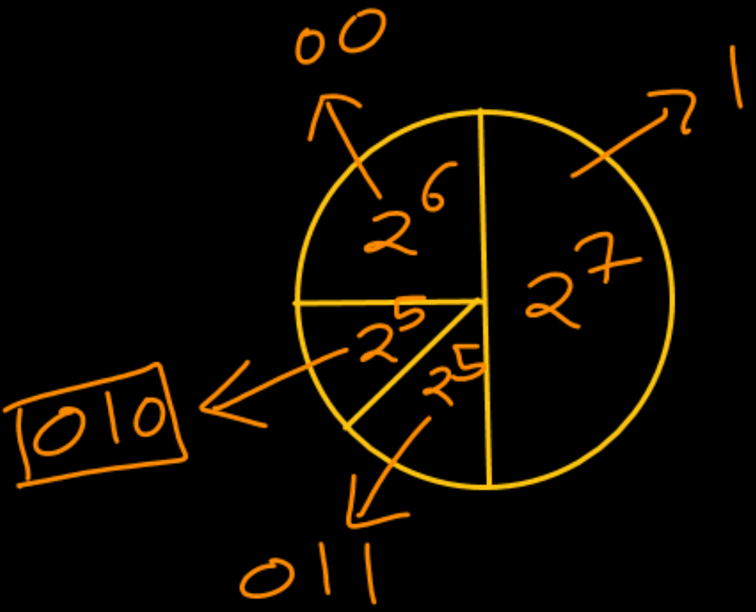
Allocated IP Address space = $\underbrace{155.220.195.\,P_1\,P_2\,P_3}_{\text{27 bit prefix}}\,\underbrace{0\,0\,0\,0\,0}_{\text{5 bit Host ID}} / 27$



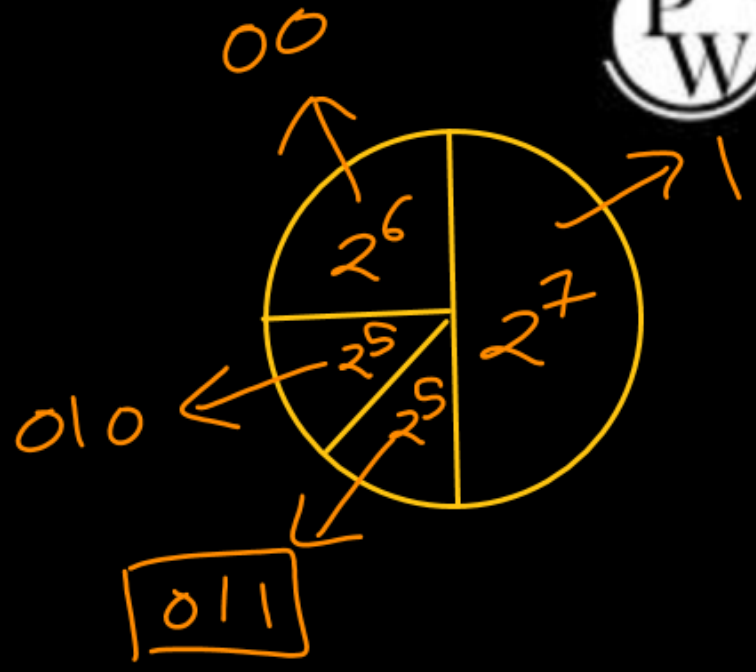
$\underbrace{000}_{P_1P_2P_3}\underbrace{00000}$



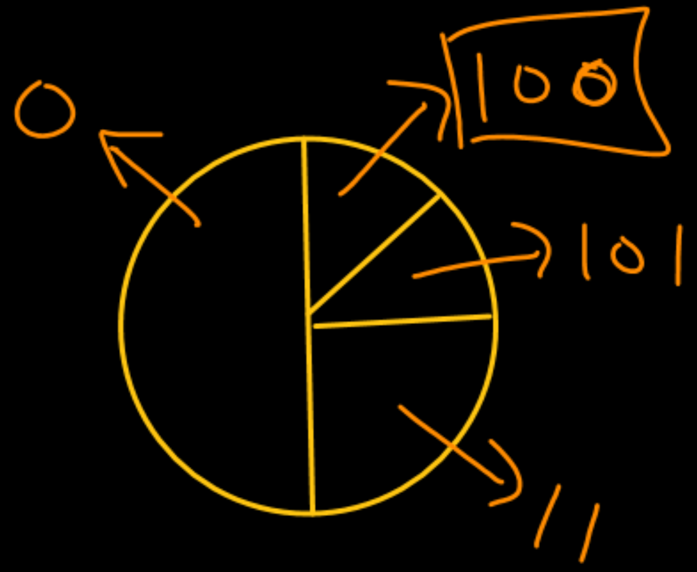
$\underbrace{001}_{P_1P_2P_3}\underbrace{00000}$



$\underbrace{010}_{P_1P_2P_3}\underbrace{00000}$



$\underbrace{011}_{P_1P_2P_3}\underbrace{00000}$



$\underbrace{100}_{P_1P_2P_3}\underbrace{00000}$

[MCQ] , H.W. IIT → [GATE-2020][2 Mark]

#Q. An organization requires a range of IP address to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space "202 . 61 . 0 . 0 / 17". The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one of the organization?

I. 202 . 61 . 84 . 0 / 21

II. 202 . 61 . 104 . 0 / 21

III. 202 . 61 . 64 . 0 / 21

IV. 202 . 61 . 144 . 0 / 21

(A) I and II only

(B) II and III only

(C) III and IV only

(D) I and IV only

[MCQ] H.W. IIT-D [GATE-2012][2 Mark]

#Q. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it : "245 . 248 . 128 . 0/ 20". The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

A) 245 . 248 . 136 . 0 / 21 and 245 . 248 . 128 . 0 / 22

B) 245 . 248 . 128 . 0 / 21 and 245 . 248 . 128 . 0 / 22

C) 245 . 248 . 132 . 0 / 22 and 245 . 248 . 132 . 0 / 21

D) 245 . 248 . 136 . 0 / 24 and 245 . 248 . 132 . 0 / 21

CIDR based available IP address space : 245 . 248 . 128 . 0 / 20

245 . 248 . 1 0 0 0 _ _ _ _ . _ _ _ _ _ _ _ _ / 20

Org.-1 Network Address : 245 . 248 .

245 . 248 . 1 0 0 0 _ _ _ _ . _ _ _ _ _ _ _ _ /

Org.-2 Network Address : 245 . 248 .

245 . 248 . 1 0 0 0 _ _ _ _ . _ _ _ _ _ _ _ _ /

ISP available Address Space : 245 . 248 .

245 . 248 . 1 0 0 0 _ _ _ _ . _ _ _ _ _ _ _ _ /

Org.-1 Net. Add. : 245 . 248 .

Org.-2 Net. Add. : 245 . 248 .

ISP Remaining : 245 . 248 .

Org.-1 Net. Add. : 245 . 248 .

Org.-2 Net. Add. : 245 . 248 .

ISP Remaining : 245 . 248 .

Org.-1 Net. Add. : 245 . 248 .

Org.-2 Net. Add. : 245 . 248 .

ISP Remaining : 245 . 248 .

Org.-1 Net. Add. : 245 . 248 .

Org.-2 Net. Add. : 245 . 248 .

ISP Remaining : 245 . 248 .

2 mins Summary

Topic Routing Table

Topic CIDR

THANK - YOU

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